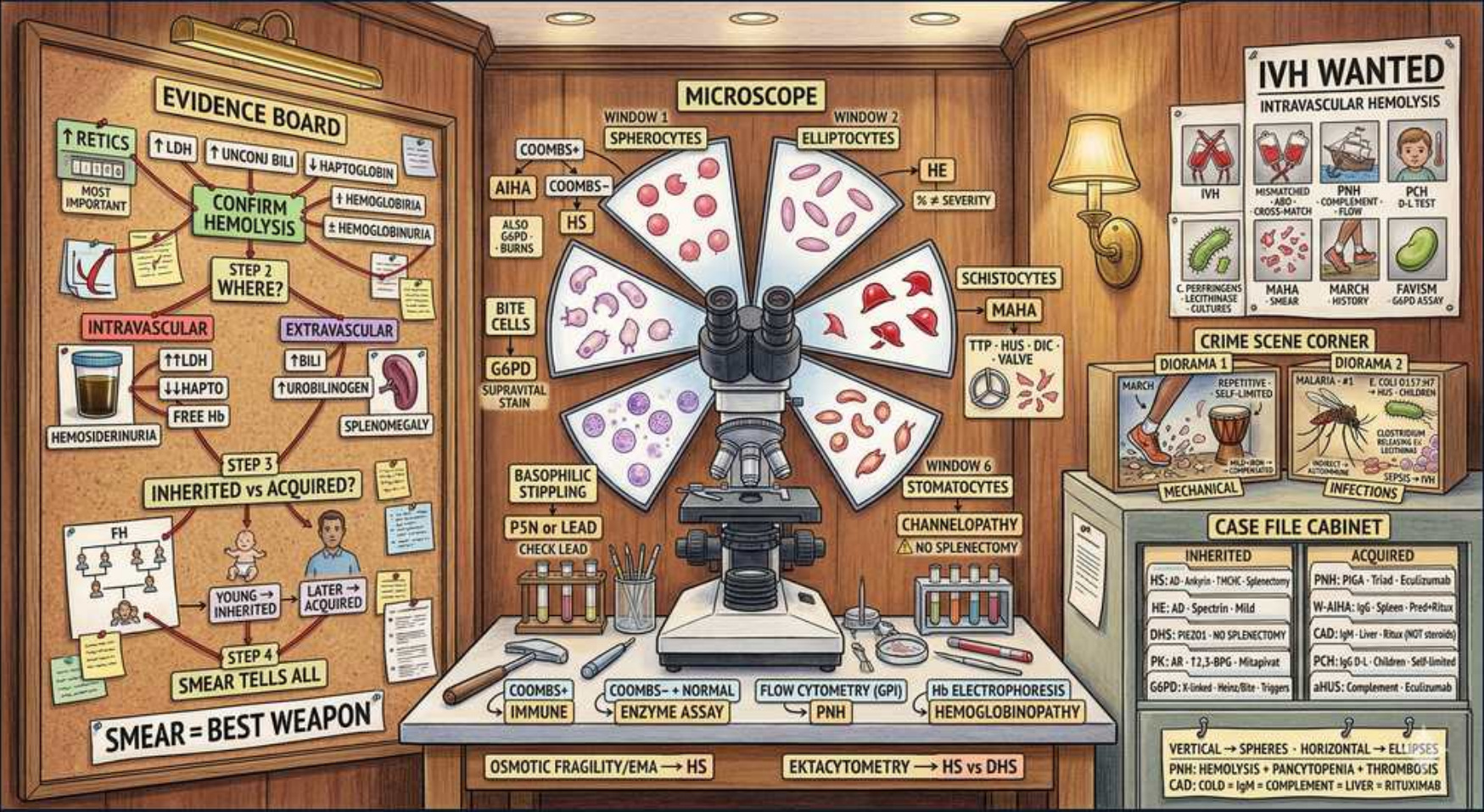


# Hemolytic Anemia: Acquired Causes & Smear Approach





## 1. Step 1: confirm hemolysis

### WHAT YOU SEE

Evidence board, LDH, haptoglobin, bilirubin

### WHAT IT ENCODES

Confirm hemolysis first: HIGH reticulocytes, HIGH LDH, HIGH indirect (unconjugated) bilirubin, LOW haptoglobin. Reticulocytosis is the most important sign of a hyperproliferative anemia. These labs establish destruction before subtyping.

### BOARD TRAP

Skipping confirmation and chasing a cause; without retics/LDH/haptoglobin you can't prove hemolysis is occurring.

## 2. Step 2: where (intra vs extra)

### WHAT YOU SEE

Intravascular vs extravascular split

### WHAT IT ENCODES

Localize: INTRAvascular = hemoglobinuria, hemosiderinuria, very low haptoglobin (PNH, MAHA, transfusion reaction, mechanical valve). EXTRAvascular = splenomegaly, spherocytes, high indirect bilirubin/urobilinogen (warm AIHA, HS). Site narrows the differential.

### BOARD TRAP

Expecting hemoglobinuria in extravascular hemolysis; free Hb stays low so urine hemoglobin signals the intravascular route.

## 3. Step 3: inherited vs acquired

### WHAT YOU SEE

Family history, young/inherited vs later/acquired

### WHAT IT ENCODES

Use age of onset and family history: lifelong hemolysis from childhood with family history suggests inherited (membrane, enzyme, hemoglobinopathy); new-onset in an adult suggests acquired (immune, MAHA, PNH, drug, infection). This branch drives testing.

### BOARD TRAP

Assuming a new adult presentation is inherited; later-onset hemolysis is usually acquired and needs DAT/smear workup.

## 4. Step 4: smear is best weapon

### WHAT YOU SEE

Banner — smear = best weapon

### WHAT IT ENCODES

The peripheral SMEAR is the highest-yield test in hemolysis: spherocytes (AIHA/HS), schistocytes (MAHA), bite cells (G6PD), sickle cells, target cells, basophilic stippling, etc. Morphology directs the next confirmatory test. Always review the smear.

### BOARD TRAP

Ordering broad reflex panels before looking at the smear; morphology focuses the workup and prevents shotgun testing.

## 5. Spherocytes

### WHAT YOU SEE

Microscope — round dense RBCs, AIHA/HS

### WHAT IT ENCODES

Spherocytes (no central pallor): warm AIHA or hereditary spherocytosis. The DAT separates them — positive in AIHA, negative in HS (confirm HS with EMA-binding flow cytometry / osmotic fragility). Both hemolyze extravascularly in the spleen.

### BOARD TRAP

Calling spherocytes diagnostic of HS; a positive DAT means warm AIHA, not hereditary spherocytosis.

## 6. Schistocytes / MAHA

### WHAT YOU SEE

Fragmented RBCs, MAHA, valve

### WHAT IT ENCODES

Schistocytes = microangiopathic hemolytic anemia (MAHA): thrombotic microangiopathies, DIC, malignant hypertension, mechanical heart valves, HELLP. Schistocytes + thrombocytopenia is a medical emergency requiring urgent evaluation. DAT negative.

### BOARD TRAP

Dismissing schistocytes with low platelets; that combination signals a thrombotic microangiopathy needing emergent workup.

## 7. Bite cells / G6PD

### WHAT YOU SEE

Bite cells, G6PD

### WHAT IT ENCODES

Bite cells and Heinz bodies (supravital stain) point to oxidative hemolysis from G6PD deficiency — triggered by infection, fava beans, or oxidant drugs (dapsone, primaquine, rasburicase). Confirm with a G6PD assay at steady state.

### BOARD TRAP

Drawing a G6PD level during the acute crisis; reticulocytes raise enzyme falsely — retest weeks later at steady state.

## 8. Elliptocytes

### WHAT YOU SEE

Window 2 — pencil/oval cells

### WHAT IT ENCODES

Elliptocytes/ovalocytes: hereditary elliptocytosis (membrane skeleton defect). Most are mild; rare severe variants resemble pyropoikilocytosis. Hereditary stomatocytosis is a related membrane disorder. Splenectomy contraindicated in some stomatocytosis (thrombosis).

### BOARD TRAP

Assuming all membrane disorders respond to splenectomy; some hereditary stomatocytoses thrombose after splenectomy.



## 9. Basophilic stippling

### WHAT YOU SEE

Stippled RBC, PSN/lead, check CBA

### WHAT IT ENCODES

Coarse basophilic stippling suggests lead poisoning or sideroblastic anemia / pyrimidine 5'-nucleotidase deficiency. Lead inhibits this enzyme, leaving ribosomal RNA aggregates. Pair with a lead level and toxic exposure history.

### BOARD TRAP

Ignoring basophilic stippling; it is a specific clue to lead toxicity or P5N deficiency, not generic hemolysis.

## 10. Coombs+ / immune

### WHAT YOU SEE

Coombs+ → immune

### WHAT IT ENCODES

A POSITIVE direct antiglobulin test means immune hemolysis: warm AIHA (IgG), cold agglutinin disease (C3), or drug-induced. Identify the antibody type to guide therapy (steroids for warm; keep-warm/rituximab for cold).

### BOARD TRAP

Treating all DAT-positive hemolysis with steroids; cold agglutinin disease is DAT(C3)+ but steroid-unresponsive.

## 11. Coombs-negative branch

### WHAT YOU SEE

Coombs normal → flow cytometry, enzyme, electrophoresis

### WHAT IT ENCODES

DAT-NEGATIVE hemolysis: pursue PNH (flow cytometry/FLAER), enzyme assays (G6PD, PK), hemoglobin electrophoresis (sickle, thalassemia), and membrane testing (EMA/osmotic fragility). The negative Coombs redirects to intrinsic and microangiopathic causes.

### BOARD TRAP

Stopping the workup at a negative DAT; that result mandates PNH, enzyme, and hemoglobinopathy testing, not reassurance.

## 12. Hb electrophoresis

### WHAT YOU SEE

Hb electrophoresis, hemoglobinopathy

### WHAT IT ENCODES

Hemoglobin electrophoresis/HPLC diagnoses hemoglobinopathies — sickle cell (HbS), thalassemias (target cells, microcytosis), HbC. These intracorporeal inherited causes hemolyze chronically with characteristic smear morphology.

### BOARD TRAP

Excluding a hemoglobinopathy without electrophoresis; sickle and thalassemia need HPLC/electrophoresis, not just smear.

## 13. IVH wanted — intravascular causes

### WHAT YOU SEE

Wanted poster, IVH causes

### WHAT IT ENCODES

Intravascular hemolysis culprits: mechanical (prosthetic valve, march hemoglobinuria), MAHA (TMA/DIC), transfusion reactions, PNH, severe infections (Clostridium, malaria, babesiosis). These show hemoglobinemia/hemoglobinuria and need urgent cause-specific treatment.

### BOARD TRAP

Forgetting infection (malaria, babesia, Clostridium) as an intravascular hemolysis cause in the right clinical setting.

## 14. March hemoglobinuria

### WHAT YOU SEE

Mechanical — march, valve

### WHAT IT ENCODES

Mechanical intravascular hemolysis: prosthetic heart valves (paravalvular leak shears RBCs, schistocytes) and march hemoglobinuria (repetitive foot trauma in runners/marchers). Both produce transient hemoglobinuria. Identify the mechanical source.

### BOARD TRAP

Working up a marathon runner's post-race hemoglobinuria as immune; march hemoglobinuria is mechanical and self-limited.

## 15. Infections (malaria/babesia)

### WHAT YOU SEE

Microbes — malaria, babesia, Clostridium

### WHAT IT ENCODES

Infections causing hemolysis: malaria (intraerythrocytic parasites, blackwater fever), babesiosis (Maltese cross), Clostridium perfringens sepsis (lecithinase lyses RBCs, spherocytes). Smear and travel/exposure history clinch these.

### BOARD TRAP

Missing babesiosis or malaria on the smear; intraerythrocytic parasites explain otherwise unexplained intravascular hemolysis.

## 16. Case file: inherited vs acquired

### WHAT YOU SEE

Case file cabinet, inherited vs acquired columns

### WHAT IT ENCODES

Final synthesis: inherited = membrane (HS, elliptocytosis), enzyme (G6PD, PK), hemoglobinopathy. Acquired = immune (AIHA), microangiopathic (TMA/DIC), PNH, infection, drugs, hypersplenism. Smear + DAT + flow cytometry assign the cause.

### BOARD TRAP

Forcing every case into one bucket; mixed pictures (e.g. PNH overlapping marrow failure) need flow cytometry plus the full panel.